

Natural Language Processing (NLP) with Python

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What is Natural Language Processing?

Computers and machines are great at working with tabular data or spreadsheets. However, as human beings generally communicate in words and sentences, not in the form of tables. Much information that humans speak or write is unstructured. So it is not very clear for computers to interpret such. In natural language processing (NLP), the goal is to make computers understand the unstructured text and retrieve meaningful pieces of information from it. Natural language Processing (NLP) is a subfield of **artificial intelligence**, in which its depth involves the interactions between computers and humans.

Applications of NLP:

- Machine Translation.
- Speech Recognition.
- Sentiment Analysis.
- Question Answering.
- Summarization of Text.
- Chatbot.
- Intelligent Systems.
- Text Classifications.
- Character Recognition.
- Spell Checking.
- Spam Detection.
- Autocomplete.
- Named Entity Recognition.
- Predictive Typing.

Understanding Natural Language Processing (NLP):

We, as humans, perform natural language processing (NLP) considerably well, but even then, we are not perfect. We often misunderstand one thing for another, and we often interpret the same sentences or words differently.

For instance, consider the following sentence, we will try to understand its interpretation in many different ways:

Example 1:

I saw a man on a hill with a telescope.

Figure 2: NLP example sentence with the text: “I saw a man on a hill with a telescope.”

These are some interpretations of the sentence shown above.

- There is a man on the hill, and I watched him with my telescope.
- There is a man on the hill, and he has a telescope.
- I’m on a hill, and I saw a man using my telescope.
- I’m on a hill, and I saw a man who has a telescope.
- There is a man on a hill, and I saw him something with my telescope.

Example 2:

Can you help me with the can?

Figure 3: NLP example sentence with the text: “Can you help me with the can?”

In the sentence above, we can see that there are two “can” words, but both of them have different meanings. Here the first “can” word is used for question formation. The second “can” word at the end of the sentence is used to represent a container that holds food or liquid.

Hence, from the examples above, we can see that language processing is not “deterministic” (the same language has the same interpretations), and something suitable to one person might not be suitable to another. Therefore, Natural Language Processing (NLP) has a non-deterministic approach. In other words, Natural Language Processing can be used to create a new intelligent system that can understand how humans understand and interpret language in different situations.

Rule-based NLP vs. Statistical NLP:

Natural Language Processing is separated in two different approaches:

Rule-based Natural Language Processing:

It uses common sense reasoning for processing tasks. For instance, the freezing temperature can lead to death, or hot coffee can burn people's skin, along with other common sense reasoning tasks. However, this process can take much time, and it requires manual effort.

Statistical Natural Language Processing:

It uses large amounts of data and tries to derive conclusions from it. Statistical NLP uses machine learning algorithms to train NLP models. After successful training on large amounts of data, the trained model will have positive outcomes with deduction.

Comparison:

Rule Based NLP	Statistical NLP
+ 1) Flexible	+ 1) Easy to scale
+ 2) Easy to debug	+ 2) Learn by itself
+ 3) Doesn't require much training	+ 3) Fast development
+ 4) Understanding language	+ 4) High coverage
+ 5) High precision	
- 1) Requires skilled developers	- 1) Requires large amount of data
- 2) Slow parsing	- 2) Difficult to debug
- 3) Moderate coverage	- 3) No understanding of context

Figure 4: Rule-Based NLP vs. Statistical NLP.

Components of Natural Language Processing (NLP):

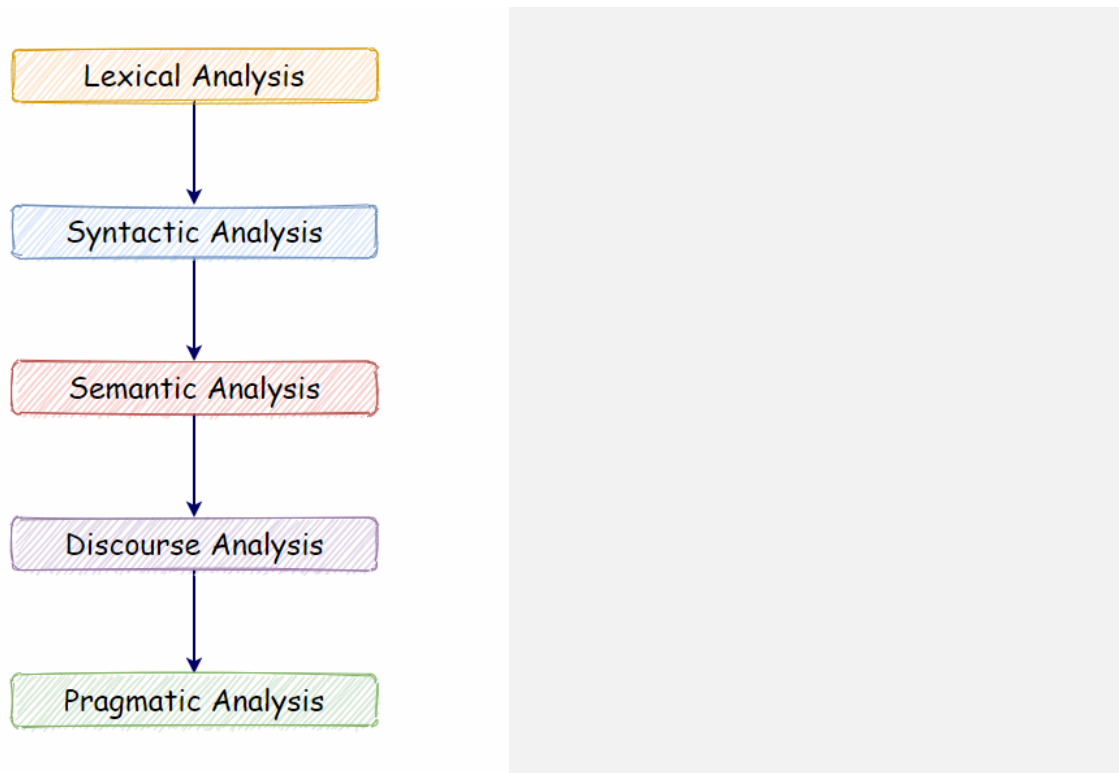


Figure 5: Components of Natural Language Processing (NLP).

a. Lexical Analysis:

With lexical analysis, we divide a whole chunk of text into paragraphs, sentences, and words. It involves identifying and analyzing words' structure.

b. Syntactic Analysis:

Syntactic analysis involves the analysis of words in a sentence for grammar and arranging words in a manner that shows the relationship among the words. For instance, the sentence "The shop goes to the house" does not pass.

c. Semantic Analysis:

Semantic analysis draws the exact meaning for the words, and it analyzes the text meaningfulness. Sentences such as "hot ice-cream" do not pass.

d. Disclosure Integration:

Disclosure integration takes into account the context of the text. It considers the meaning of the sentence before it ends. For example: "He works at Google." In this sentence, "he" must be referenced in the sentence before it.

e. Pragmatic Analysis:

Pragmatic analysis deals with overall communication and interpretation of language. It deals with deriving meaningful use of language in various situations.

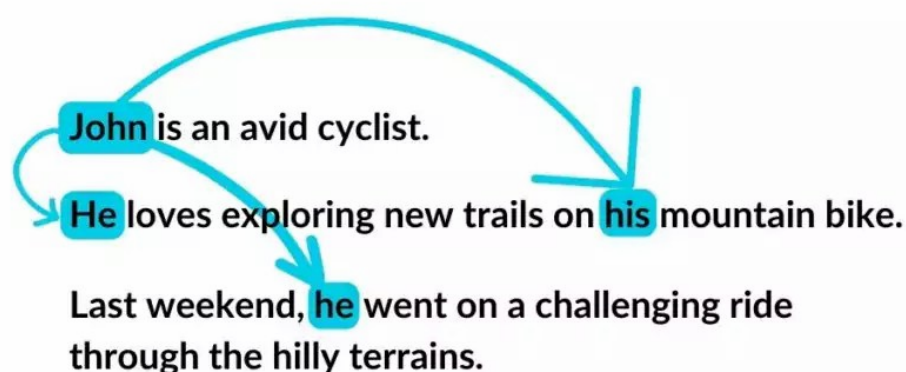
Current challenges in NLP:

1. Breaking sentences into tokens.
2. Tagging parts of speech (POS).
3. Building an appropriate vocabulary.
4. Linking the components of a created vocabulary.
5. Understanding the context.
6. Extracting semantic meaning.
7. Named Entity Recognition (NER).
8. Transforming unstructured data into structured data.
9. Ambiguity in speech.

Coherence Resolution

Coherence resolution is a crucial aspect of Natural Language Processing (NLP) that involves identifying and linking expressions in text that refer to the same entity. In other words, it determines when two or more words or phrases in a text document refer to the same thing or person. This process is essential for machines to understand and interpret text coherently, as humans naturally grasp the connections between pronouns, names, and other referring expressions.

Consider the following example:



“John is an avid cyclist. He loves exploring new trails on his mountain bike. Last weekend, he went on a challenging ride through the hilly terrains.”

In this case, coreference resolution helps the system understand that “He” and “his” in the second sentence and “he” in the third sentence refer to “John” in the first sentence. This understanding is crucial for tasks such as document [summarisation](#), [question answering](#), and [machine translation](#), where maintaining context and coherence is essential.

Easy to use NLP libraries:

a. **NLTK (Natural Language Toolkit):**

The NLTK Python framework is generally used as an education and research tool. It’s not usually used on production applications. However, it can be used to build exciting programs due to its ease of use.

Features:

- Tokenization.
- Part Of Speech tagging (POS).
- Named Entity Recognition (NER).
- Classification.
- Sentiment analysis.
- Packages of chatbots.

Use-cases:

- Recommendation systems.
- Sentiment analysis.
- Building chatbots.

Pros	Cons
+ 1) Most well known and full NLP library	- 1) Difficult to learn and use
+ 2) Supports largest numbers of languages	- 2) Context of word is ignored
	- 3) Slow
	- 4) No neural network model

Figure 6: Pros and cons of using the NLTK framework.

b. **spaCy:**

spaCy is an open-source natural language processing Python library designed to be fast and production-ready. spaCy focuses on providing software for production usage.

Features:

- Tokenization.
- Part Of Speech tagging (POS).
- Named Entity Recognition (NER).
- Classification.
- Sentiment analysis.
- Dependency parsing.
- Word vectors.

Use-cases:

- Autocomplete and autocorrect.
- Analyzing reviews.
- Summarization.

Pros	Cons
+ 1) Fast	- 1) Less flexible
+ 2) Easy to learn and use	
+ 3) Use neural networks for training	

Figure 7: Pros and cons of the spaCy framework.

c. Gensim:

Gensim is an NLP Python framework generally used in topic modeling and similarity detection. It is not a general-purpose NLP library, but it handles tasks assigned to it very well.

Features:

- Latent semantic analysis.
- Non-negative matrix factorization.
- TF-IDF.

Use-cases:

- Converting documents to vectors.
- Finding text similarity.
- Text summarization.

Pros	Cons
+ 1) Intuitive interface	- 1) Designed for unsupervised text models
+ 2) Scalable	- 2) Should be used with other libraries
+ 3) Implemented algorithms	

Figure 8: Pros and cons of the Gensim framework.

d. **Pattern:**

Pattern is an NLP Python framework with straightforward syntax. It's a powerful tool for scientific and non-scientific tasks. It is highly valuable to students.

Features:

- Tokenization.
- Part of Speech tagging.
- Named entity recognition.
- Parsing.
- Sentiment analysis.

Use-cases:

- Spelling correction.
- Search engine optimization.
- Sentiment analysis.

Pros	Cons
+ 1) Data mining	- 1) Not optimized with specific NLP tasks
+ 2) Network analysis and visualization	

Figure 9: Pros and cons of the Pattern framework.

e. **TextBlob:**

TextBlob is a Python library designed for processing textual data.

Features:

- Part-of-Speech tagging.
- Noun phrase extraction.
- Sentiment analysis.
- Classification.
- Language translation.

- Parsing.
- Wordnet integration.

Use-cases:

- Sentiment Analysis.
- Spelling Correction.
- Translation and Language Detection.

Pros	Cons
+ 1) Easy to use	- 1) Slow
+ 2) Intuitive interface to NLTK	- 2) No neural network model
+ 3) Provides language translation and detection	- 3) No integrated word vectors

Figure 10: Pros and cons of the TextBlob library.

